
Bioscara

Sebastian Storz

Feb 04, 2026

CONTENTS:

1	Overview	1
1.1	Documentation	1
1.2	User Guides	33
	Index	39

OVERVIEW

TODO:

Note

A project overview should briefly describe the scope, purpose, and goals of the project. It helps users quickly understand what the project is about and why it exists. When writing an overview, mention the main problem the project solves, its target audience, and any unique features or technologies it uses.

Example:

This project provides a standardized template for DALSA GitHub repositories, designed to help project owners quickly set up new projects with a consistent structure and DevOps features. Its goal is to ensure maintainability and collaboration by enforcing standards that make future contributions and usage seamless. The template integrates Sphinx + Doxygen for unified documentation of both Python and C++ packages, and uses GitHub Actions to automate generation of web-based documentation, code analysis (linting), code rewrites to follow a consistent style (formatting), and execution of tests.

Here you will find a full description of the project's architecture and core concepts, detailed documentation of all packages, and practical usage examples. This guide is designed to help you understand the structure and functionality of the project, making it easier to get started and make effective use of its features.

TODO:

1.1 Documentation

The C++ API documentation is generated with doxygen and can be found [here](#).

1.1.1 bioscara_hardware_interfaces

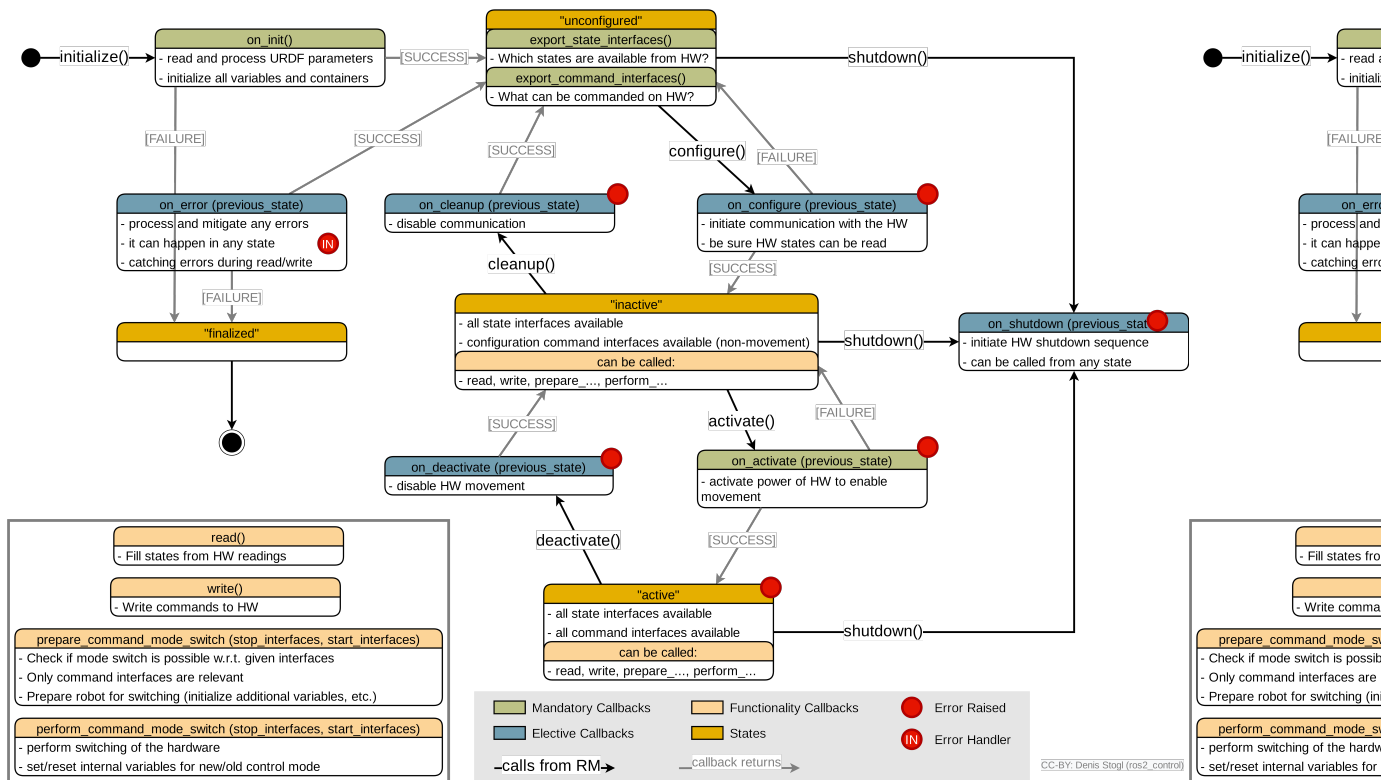
```
constexpr char bioscara_hardware_interfaces::HW_IF_HOME[] = "home"
```

```
class BioscaraArmHardwareInterface : public hardware_interface::SystemInterface
```

```
#include <arm_hardware.hpp> The bioscara arm hardware interface class.
```

The hardware interface serves to wrap custom hardware interaction with the arm joints in the standardized ros2_control architecture.

Hardware Lifecycle The hardware follows the ros2_control hardware interface lifecycle which internally is following the [ROS2 managed node lifecycle](#).



Public Functions

hardware_interface::CallbackReturn **on_init**(const hardware_interface::HardwareComponentInterfaceParams ¶ms) override

Called on initialization to the **unconfigured** state.

Performs the following checks on the configures joints parsed from the URDF description:

- Each joint must have the 3 command interfaces (in this order): 'position', 'velocity', 'home'
- Each joint must have the 3 state interfaces (in this order): 'position', 'velocity', 'home'

Stores the configuration parameters for each joint in the `_joint_cfg` map. Each joint must have these parameters:

- `i2c_address` (int, HEX)
- `reduction` (float)
- `min` (float)
- `max` (float)
- `stall_threshold` (int, DEC)
- `hold_current` (int, DEC)
- `drive_current` (int, DEC)
- `max_acceleration` (float)
- `max_velocity` (float)
- `homing`

- speed (float)
- threshold (int, DEC)
- current (int, DEC)
- acceleration (float)

Adds each joint to the internal `_joints` map. Creates a `MockJoint` object if the `use_mock_hardware` parameter is 'True' or 'true', or else a hardware Joint.

Parameters

params –

Returns

`hardware_interface::CallbackReturn`

`hardware_interface::CallbackReturn` **on_shutdown**(const `rcpp_lifecycle::State` &previous_state) override

Called on the transition from the `inactive`, `unconfigured` and `active` to the `finalized` state.

When transitioning directly from `active` to `finalized` `on_deactivate()` is automatically called before [Source Code](#) If the previous state is either `inactive` or `active` the `on_cleanup()` method is called first. Then regardless of the previous state, the `_joints` map is cleared.

Parameters

previous_state –

Returns

`hardware_interface::CallbackReturn`

`hardware_interface::CallbackReturn` **on_configure**(const `rcpp_lifecycle::State` &previous_state) override

Called on the transition from the `unconfigured` to the `inactive` state.

Establish and test connection to each joint.

Parameters

previous_state –

Returns

`hardware_interface::CallbackReturn`

`hardware_interface::CallbackReturn` **on_cleanup**(const `rcpp_lifecycle::State` &previous_state) override

Called on the transition from the `inactive` to the `unconfigured` state.

Disconnect from the joints.

Parameters

previous_state –

Returns

`hardware_interface::CallbackReturn`

`hardware_interface::CallbackReturn` **on_activate**(const `rcpp_lifecycle::State` &previous_state) override

Called on the transition from the `inactive` to the `active` state.

Calls `activate_joint()` to enable the joints.

It is allowed to activate the hardware even if it is not homed. To home the joint the homing_controller must be activated, but generally a hardware component must be active in order for controllers to become active.

To prohibit movement on activation the set point for each position command interface is set equal to the current measured position, and the velocity command is set to 0.0 for each command interface. The current values are obtained by calling the `read()` method once which populates the state interfaces with values.

Parameters

previous_state –

Returns

hardware_interface::CallbackReturn

hardware_interface::CallbackReturn **on_deactivate**(const rclcpp_lifecycle::State &previous_state) override

Called on the transition from the `active` to the `inactive` state.

Disables all joints and thereby allows backdriving. State interfaces continue to be updated.

Parameters

previous_state –

Returns

hardware_interface::CallbackReturn

hardware_interface::return_type **read**(const rclcpp::Time &time, const rclcpp::Duration &period) override

Reads from the hardware and populates the state interfaces.

Iterates over all state interfaces and calls the corresponding Joint method.

- State interface “position” -> *bioscara_hardware_drivers::Joint::getPosition()*
 - State interface “velocity” -> *bioscara_hardware_drivers::Joint::getVelocity()*
 - State interface “home” -> *bioscara_hardware_drivers::Joint::isHomed()*
 - This does not actually trigger a communication, instead it relies on the return flags of the previous transmissions. Since position and velocity have been called immediately before the return flags are assumed to be valid.
 - If the homing of a joint has been activated through the command interface (*bioscara_hardware_drivers::Joint::getCurrentBCmd() == bioscara_hardware_drivers::Joint::HOME*) the device signals BUSY (*bioscara_hardware_drivers::Joint::isBusy()*) as long as it is still homing.
- If the BUSY flag is reset while the current command is still *bioscara_hardware_drivers::Joint::HOME* we can assume the homing has finished. Then the “home” command interface of the joint is reset to 0.0, which will stop the homing (perform cleanup tasks) at the next write cycle.

Parameters

- **time** –
- **period** –

Returns

hardware_interface::return_type

hardware_interface::return_type **write**(const rclcpp::Time &time, const rclcpp::Duration &period) override

Writes commands to the hardware from the command interfaces.

In contrast to the *read()* method the *write()* method only loops over the command interfaces that are currently active defined by the *_joint_command_modes* map. See *prepare_command_mode_switch()* for a detailed reasoning why this approach has been chosen.

- Command interface “position” -> *bioscara_hardware_drivers::Joint::setPosition()*

- Command interface “velocity” -> `bioscara_hardware_drivers::Joint::setVelocity()`
- Command interface “home” -> `bioscara_hardware_drivers::Joint::startHoming()`
 - If the commanded value in “home” is $\neq 0.0$ and the joint is currently executing a blocking function, for example homing (`bioscara_hardware_drivers::Joint::getCurrentBCmd() == bioscara_hardware_drivers::Joint::NONE`), the homing sequence is started with the speed, sensitivity, current and acceleration defined in the `_joint_cfg` which is populated from the hardware description urdf. The direction of the homing is determined by the sign of the command interface value.
 - If the commanded value in “home” is $= 0.0$ and the joint is currently executing homing, the homing is stopped. This can either happen prematurely through user input or when the homing is completed which is registered in `read()`.

Parameters

- **time** –
- **period** –

Returns

hardware_interface::return_type

hardware_interface::return_type **prepare_command_mode_switch**(const std::vector<std::string> &start_interfaces, const std::vector<std::string> &stop_interfaces) override

Performs checks and book keeping of the active control mode when changing controllers in non-realtime context.

For safe operation only one controller may interact with the hardware at the time. For example if the velocity JTC is active and has claimed the velocity command interfaces it is technically possible to activate the position JTC (or a homing controller, or others) that claim a different command interface (position in this case). However if both controllers are active they start writing to the hardware simultaneously which is to be avoided. For this reason a book keeping mechanism has been implemented which stores the currently active command interfaces for each joint in the `_joint_command_modes` member. Each joint has a set of active command interfaces. When a controller switch is performed the interfaces that should be stopped are removed from each joint set, then the one that should be started are added, if they are already present an error is thrown. Lastly a validation is performed. Currently the validation is simple since each joint may only have one command interface. The validation can be expanded for future use cases that require a combination of active command interfaces per joint for example.

The following basic checks are implemented:

- **On deactivation:**
 - [ERROR] Homing command interfaces may only be deactivated if no current homing process is ongoing (`bioscara_hardware_drivers::Joint::getCurrentBCmd() != bioscara_hardware_drivers::Joint::HOME`)
 - [WARN] Deactivating a velocity command interface if the velocity set point is 0.0.
 - [WARN] Deactivating a command interface that has not been started. This should not happen.
- **On activation:**
 - [ERROR] Activating a command interface that is already started. This should not happen.
 - [ERROR] Activating a second command interface for a joint.

- [ERROR] Activating ‘position’ or ‘velocity’ command interface if the joint is not homed (*bioscara_hardware_drivers::Joint::isHomed()* == false).

Since this method operates in non-realtime context it must not access critical members (*_joint_command_modes* and *_joints*) to avoid priority inversion. Therefore the new command modes are first saved to a cache *_new_joint_command_modes*. This will then be applied to *_joint_command_modes* in the *perform_command_mode_switch()* which is executed in RT context.

Parameters

- **start_interfaces** – command interfaces that should be started in the form “joint/interface”
- **stop_interfaces** – command interfaces that should be stopped in the form “joint/interface”

Returns

hardware_interface::return_type

hardware_interface::return_type **perform_command_mode_switch**(const std::vector<std::string> &start_interfaces, const std::vector<std::string> &stop_interfaces) override

Perform the mode-switching for the new command interface combination in realtime context.

Performs the following actions:

- acquires the *mtx* lock to protect the critical members.
- Copies the cached *_new_joint_command_modes* to the *_joint_command_modes*
- **On activation:**
 - **home** interface:
 - * Reset command to 0.0. This clears any remaining commands that have been written to the command interface while the hardware was unable to act on it. For example if it was inactive or the homing command was not the active command mode.

Parameters

- **start_interfaces** – vector of string identifiers for the command interfaces starting.
- **stop_interfaces** – vector of string identifiers for the command interfaces stopping.

Returns

return_type::OK if the new command interface combination can be switched to (or) if the interface key is not relevant to this system. Returns return_type::ERROR otherwise.

hardware_interface::CallbackReturn **on_error**(const rclcpp_lifecycle::State &previous_state) override

Called when an error in any state or state transition is thrown.

According to the [ros2_control documentation](#):

Error handling follows the node lifecycle. If successful CallbackReturn::SUCCESS is returned and hardware is again in UNCONFIGURED state, if any ERROR or FAILURE happens the hardware ends in FINALIZED state and can not be recovered. The only option is to reload the complete plugin, but there is currently no service for this in the Controller Manager.

Since the hardware will immediatly return to the *unconfigured* state ([source](#)) if the error could be handled we manually call the transition functions which would normally be called to this state. Those are:

- **Previous state:** active
 - Deactivate hardware (*on_deactivate()*) -> inactive
 - Clean-Up hardware (*on_cleanup()*) -> unconfigured
- **Previous state:** inactive
 - Deactivate hardware (*on_deactivate()*) -> inactive
 - * call the deactivate function anyway regardless if state was active or inactive. For example if the *on_activate()* function fails on *bioscara_hardware_drivers::Joint::enableStallguard()* the joint will have been enabled, to disable it invoke *on_deactivate()*.
 - Clean-Up hardware (*on_cleanup()*) -> unconfigured

In particular the deactivation is important. For example if a joint stalls the *read()* or *write()* methods throw an error, which will be handled here and allow the hardware to be deactivated, disabling the joints to allow backdriving.

Parameters

previous_state –

Returns

hardware_interface::CallbackReturn

Private Functions

bioscara_hardware_drivers::err_type_t **start_homing**(const std::string name, float velocity)

wrapper method to start homing.

Activate the joint, set homing acceleration and start homing.

Parameters

- **name** –
- **velocity** –

Returns

bioscara_hardware_drivers::err_type_t

bioscara_hardware_drivers::err_type_t **stop_homing**(const std::string name)

wrapper method to stop homing.

Stop the homing. Reset acceleration and velocity and perform the postHoming cleanup, then deactivate the joint.

Parameters

name –

Returns

bioscara_hardware_drivers::err_type_t

void **split_interface_string_to_joint_and_name**(std::string interface, std::string &joint_name, std::string &interface_name)

Split a interface string like “<joint_name>/<interface_name>” to “<joint_name>” and “<interface_name>”.

Parameters

- **interface** –
- **joint_name** –

- **interface_name** –

`bioscara_hardware_drivers::err_type_t activate_joint(const std::string name)`

Enables each joint, enables the stall detection and sets the maximum acceleration.

Parameters

name – joint name to enable

Returns

bioscara_hardware_drivers::err_type_t

`bioscara_hardware_drivers::err_type_t deactivate_joint(const std::string name)`

Disables each joint.

Parameters

name – joint name to disable

Returns

bioscara_hardware_drivers::err_type_t

Private Members

`std::unordered_map<std::string, std::unique_ptr<bioscara_hardware_drivers::BaseJoint>> _joints`

unordered map storing the pointers to BaseJoint objects. This will either be a MockJoint or Joint.

An unordered map is chosen to simplify access via the joint name, as this conforms well with the ROS2_control hardware interface. The map does not need to be ordered. Search, insertion, and removal of elements have average constant-time complexity.

Since the BaseJoint methods are implemented as virtual, dynamic method dispatch can be utilized to call the correct implementation of a method. So either BaseJoint::foo() or Joint::foo()/MockJoint::foo() if foo() is overwritten in Joint or MockJoint. A smart pointer is used to guarantee destruction when the pointer is destructed. A unique pointer is used to prevent copying of the object.

`std::unordered_map<std::string, joint_config_t> _joint_cfg`

unordered map storing the configuration struct of the joints.

An unordered map is chosen to simplify access via the joint name, as this conforms well with the ROS2_control hardware interface. The map does not need to be ordered. Search, insertion, and removal of elements have average constant-time complexity.

`std::unordered_map<std::string, std::set<std::string>> _joint_command_modes`

unordered map of sets storing the active command interfaces for each joint.

Each joint can have a set of active command interfaces. This type of structure is chosen to group interfaces by joint. In the *write()* function the interface name can simply be constructed by concatenating joint name with interface name. Although currently only one active command interface is allowed at the time, a set can be used to store multiple command interfaces that are acceptable to be combined, for example it would be acceptable to set velocity and driver current and hence that would be an allowable combination.

An unordered map is chosen to simplify access via the joint name, as this conforms well with the ROS2_control hardware interface. The map does not need to be ordered. Search, insertion, and removal of elements have average constant-time complexity.

`std::unordered_map<std::string, std::set<std::string>> _new_joint_command_modes`

Temporary cache of new joint_command_modes when switching controllers.

Since the *prepare_command_mode_switch()* is executed in a non-RT context we save the new joint command modes to this cache first to avoid needing to lock the *_joint_command_modes*.

`std::vector<std::pair<std::string, hardware_interface::InterfaceDescription*>>`

`_ordered_joint_state_interfaces_ptr`

A vector of a pair of interface name and pointer of the hardware's state interface which is ordered by joint and state interface type.

A vector is chosen since it guarantees the correct order by insertion. The vector holds pointers to the actual interface structs stored in the parents `HardwareComponentInterface::joint_state_interfaces_` but in the desired order. The order is:

- i. position
- ii. velocity
- iii. home

This order guarantees that when reading the state interfaces in *read()* that the 'home' interface is read last and has the latest state flags from the joint.

`std::mutex` **`mtx`**

A mutex that is used to prevent concurrent access to hardware and *_joint_command_modes*.

The mutex prevents two things:

- Modifying the *_joint_command_modes* concurrently.
- Concurrent access to the hardware via the *_joints* map. The Joint hardware is not thread safe. In particular the *read()* and *write()* methods are executed in one RT thread while the *perform_command_mode_switch()* is called from another RT thread. The latter also tries to modify the Joint object via *activate_joint()* and *deactivate_joint()* which must not happen concurrently with a *read()* or *write()* call.

All methods that need to acquire this lock need to be performed in RT context to avoid being preempted by other lower priority threads potentially blocking the other RT threads from continuing.

`struct` **`joint_config_t`**

configuration structure holding the passed parameters from the `ros2_control` urdf

Saving all parameters on initialization in a structure allows for quick access during runtime.

Public Members

`int` **`i2c_address`**

[int, HEX] I2C device address

1-byte I2C device address (0x11 ... 0x14) for J1 ... J4

`float` **`reduction`** = 1

[float] Gear reduction of the joint

This is used to transform position and velocity values between in joint units and actuator (stepper) units. The sign depends on the direction the motor is mounted and is turning. Adjust such that the joint moves in the positive direction on positive joint commands. Cable polarity has no effect since the motors automatically adjust to always run in the 'right' direction from their point of view.

J1: 35

J2: $-2\pi/0.004$ (4 mm linear movement per stepper revolution, positive rotation makes the joint go down (negative), hence the minus sign for correction)

J3: 24

J4: 12

float **min**

[float] Lower joint limit in joint units

Initial values are (subject to tuning)

J1: -3.04647

J2: -0.0016

J3: -2.62672

J4: -3.01069

float **max**

[float] Upper joint limit in joint units

Initial values are (subject to tuning)

J1: 3.04647

J2: 0.3380

J3: 2.62672

J4: 3.01069

u_int8_t **drive_current**

[int, DEC] Drive current of stepper driver in 0-100 % of 2.5A output (check uStepper doc.)

Set when joint is enabled, see [*bioscara_hardware_drivers::BaseJoint::enable\(\)*](#).

u_int8_t **hold_current**

[int, DEC] Hold current of stepper driver in 0-100 % of 2.5A output (check uStepper doc.)

Set when joint is enabled, see [*bioscara_hardware_drivers::BaseJoint::enable\(\)*](#).

u_int8_t **stall_threshold**

[int, DEC] Stall protection threshold in 0-255, where lower is more sensitive.

If the PID error exceeds the set threshold a stall is triggered and the motor disabled. Set when joint is enabled, see [*bioscara_hardware_drivers::BaseJoint::enableStallguard\(\)*](#).

float **max_velocity**

[float] Maximum permitted joint velocity.

In rad/s or m/s for cylindrical and prismatic joints respectively. Set when joint is enabled, see [*bioscara_hardware_drivers::BaseJoint::setMaxVelocity\(\)*](#).

Note

This is the “last” limit and shall ALWAYS be greater than all limits set in the controller and application (MoveIt) configuration.

float **max_acceleration**

[float] Maximum permitted joint acceleration (and deceleration).

In rad/s^2 or m/s^2 for cylindrical and prismatic joints respectively. Set when joint is enabled, see [*bioscara_hardware_drivers::BaseJoint::setMaxAcceleration\(\)*](#).

Note

This is the “last” limit and shall ALWAYS be greater than all limits set in the controller and application (MoveIt) configuration.

[*joint_homing_config_t*](#) **homing**

Holding the [*joint_homing_config_t*](#) configuration values.

struct **joint_homing_config_t**

configuration structure holding the passed homing paramters from the ros2_control urdf

Saving all parameters on initialization in a structure allows for quick access during runtime. See [*bioscara_hardware_drivers::BaseJoint::_home\(\)*](#) for the description of the homing procedure.

Public Members

float **speed**

[float] Signed homing velocity.

In rad/s or m/s for cylindrical and prismatic joints respectively. Must satisfy: $1.0 < \text{RAD2DEG}(\text{JOINT2ACTUATOR}(<\text{speed}>, \text{reduction}, 0)) / 6 < 250.0$

u_int8_t **threshold**

[int, DEC] Encoder error threshold 0 to 255.

Lower values make the homing more sensitive but also more susceptible to false positives.

u_int8_t **current**

[int, DEC] Homing current between 0 and 100.

Determines how easy it is to stop the motor and thereby provoke a stall. See [*joint_config_t::drive_current*](#).

float **acceleration** = 0.01

[float] Joint acceleration during homing.

In rad/s^2 or m/s^2 for cylindrical and prismatic joints respectively. Accelerating too fast can trigger a stall and hence false positive homing. See [*joint_config_t::max_acceleration*](#).

class **BioscaraGripperHardwareInterface** : public hardware_interface::SystemInterface

#include <gripper_hardware.hpp> The bioscara gripper hardware interface class.

System interface has been chosen to allow future modifications for grippers that provide feedback. refer to the [*BioscaraArmHardwareInterface*](#) class for a more detailed description of the hardware life-cycle.

Public Functions

hardware_interface::CallbackReturn **on_init**(const
hardware_interface::HardwareComponentInterfaceParams
¶ms) override

hardware_interface::CallbackReturn **on_shutdown**(const rclcpp_lifecycle::State &previous_state) override

Called on the transition from the *inactive*, *unconfigured* and *active* to the *finalized* state.

When transitioning directly from *active* to *finalized* [*on_deactivate\(\)*](#) is automatically called before [Source Code](#) If the previous state is either *inactive* or *active* the [*on_cleanup\(\)*](#) method is called first.

Parameters
previous_state –

Returns
hardware_interface::CallbackReturn

hardware_interface::CallbackReturn **on_configure**(const rclcpp_lifecycle::State &previous_state) override

Called on the transition from the *unconfigured* to the *inactive* state.

Parameters
previous_state –

Returns
hardware_interface::CallbackReturn

hardware_interface::CallbackReturn **on_cleanup**(const rclcpp_lifecycle::State &previous_state) override

Called on the transition from the *inactive* to the *unconfigured* state.

Parameters
previous_state –

Returns
hardware_interface::CallbackReturn

hardware_interface::CallbackReturn **on_activate**(const rclcpp_lifecycle::State &previous_state) override

Called on the transition from the *inactive* to the *active* state.

Enables PWM generation.

Parameters
previous_state –

Returns
hardware_interface::CallbackReturn

hardware_interface::CallbackReturn **on_deactivate**(const rclcpp_lifecycle::State &previous_state) override

Called on the transition from the *active* to the *inactive* state.

Disables PWM generation.

Parameters
previous_state –

Returns

hardware_interface::CallbackReturn

hardware_interface::return_type **read**(const rclcpp::Time &time, const rclcpp::Duration &period) override

Reads from the hardware and populates the state interfaces.

Invokes the [bioscara_hardware_drivers::Gripper::getPosition\(\)](#) method to read the gripper position.**Parameters**

- **time** –
- **period** –

Returns

hardware_interface::return_type

hardware_interface::return_type **write**(const rclcpp::Time &time, const rclcpp::Duration &period) override

Writes commands to the hardware from the command interfaces.

Invokes the [bioscara_hardware_drivers::Gripper::setPosition\(\)](#) method to write the command to the gripper.**Parameters**

- **time** –
- **period** –

Returns

hardware_interface::return_type

hardware_interface::CallbackReturn **on_error**(const rclcpp_lifecycle::State &previous_state) override

Called when an error in any state or state transition is thrown.

According to the [ros2_control documentation](#):

Error handling follows the node lifecycle. If successful CallbackReturn::SUCCESS is returned and hardware is again in UNCONFIGURED state, if any ERROR or FAILURE happens the hardware ends in FINALIZED state and can not be recovered. The only option is to reload the complete plugin, but there is currently no service for this in the Controller Manager.

Since the hardware will immediatly return to the `unconfigured` state ([source](#)) if the error could be handled we manually call the transition functions which would normally be called to this state. Those are:

- **Previous state: active**
 - Deactivate hardware ([on_deactivate\(\)](#)) -> `inactive`
 - Clean-Up hardware ([on_cleanup\(\)](#)) -> `unconfigured`
- **Previous state: inactive**
 - Deactivate hardware ([on_deactivate\(\)](#)) -> `inactive`
 - * call the deactivate function anyway regardless if state was active or inactive.
 - Clean-Up hardware ([on_cleanup\(\)](#)) -> `unconfigured`

Parameters**previous_state** –**Returns**

hardware_interface::CallbackReturn

Private Members

`std::unique_ptr<bioscara_hardware_drivers::BaseGripper> _gripper`

Smart pointer to the local BaseGripper.

This will be used to either interact with the hardware or mock hardware. A smart pointer is used to guarantee destruction when the pointer is destructed. A unique pointer is used to prevent copying of the object.

gripper_config_t `_gripper_cfg`

configuration struct of the gripper.

`float _last_pos = std::numeric_limits<double>::quiet_NaN()`

`float _vel = std::numeric_limits<double>::quiet_NaN()`

`struct gripper_config_t`

configuration structure holding the passed paramters from the ros2_control urdf

Saving all parameters on initialization in a structure allows for quick access during runtime.

Public Members

`float reduction = 1`

`float offset = 0`

`float min`

`float max`

`float init_pos`

1.1.2 bioscara_hardware_drivers

`enum class bioscara_hardware_drivers::err_type_t`

Enum defining common error types.

Values:

enumerator **OK**

Success.

enumerator **ERROR**

Generic Error.

enumerator **NOT_HOMED**

Joint not homed or failed to home.

enumerator **NOT_ENABLED**

Joint not enabled or failed to enable.

enumerator **STALLED**

Joint stalled.

enumerator **NOT_INIT**

Joint not initialized.

enumerator **COMM_ERROR**

Communication error.

enumerator **INVALID_ARGUMENT**

Argument violates conditions.

enumerator **INCORRECT_STATE**

Joint is busy executing another blocking command.

std::string bioscara_hardware_drivers::error_to_string(err_type_t err)

Converts an error code to a string and returns it.

Parameters

err –

Returns

std::string

class **BaseGripper**

#include <mBaseGripper.h> Generic base class for gripper control implementations.

This class is a wrapper function to interact with the robot gripper either through a *MockGripper* or hardware *Gripper* object.

Subclassed by *bioscara_hardware_drivers::Gripper*, *bioscara_hardware_drivers::MockGripper*

Public Functions

BaseGripper(float reduction, float offset, float min, float max, float backup_init_pos)

Construct a new *BaseGripper* object.

The gripper has the reduction r and offset o parameters which are used to translate from a desired gripper width to the servo angle. The relationship between gripper width w and actuator angle α is as follows:

$$\alpha = r(w - o)$$

To determine these parameters execute the following steps:

- i. Manually set the gripper to an open position by setting a actuator angle. Be careful to not exceed the physical limits of the gripper since the actuator is strong enough to break PLA before stalling.
- ii. Measure the gripper width w_1 and note the set actuator angle α_1 .

- iii. Move the gripper to a more closed position that still allows you to accurately measure the width
- iv. Measure the second width w_2 and note the corresponding angle α_2
- v. Calculate the offset o :

$$o = \frac{\alpha_1 w_2 - \alpha_2 w_1}{\alpha_1 - \alpha_2}$$

- vi. Calculate the reduction r :

$$r = \frac{\alpha_1}{w_1 - o}$$

Parameters

- **reduction** – the gripper width to actuator reduction ratio
- **offset** – gripper width to actuator zero offset
- **min** – lower limit in m
- **max** – upper limit in m
- **backup_init_pos** – initial position assumed if none can be retrieved from the buffer file

~BaseGripper(void)

Destroy the *BaseGripper* object.

invokes the *disable()* and *deinit()* method to clean up.

virtual *err_type_t* **init**(void)

Initializes the gripper.

Returns

err_type_t::OK

virtual *err_type_t* **deinit**(void)

Deinitializes the gripper.

Returns

err_type_t::OK

virtual *err_type_t* **enable**(void)

Enables the gripper.

Since the gripper has no position feedback its position is initially unknown. For this reason we try to retrieve the latest commanded position from a buffer file using *retrieve_last_position()*. If this fails the *_backup_init_pos* is used.

Returns

err_type_t::OK

virtual *err_type_t* **disable**(void)

Disables the gripper.

As described in *enable()*

Returns

err_type_t::OK

virtual *err_type_t* **setPosition**(float width)

Sets the gripper width in m.

Arguments outside the allowed range are bounded to `_min` and `_max`.

Parameters

width – width in m.

Returns

err_type_t::OK

virtual *err_type_t* **getPosition**(float &width)

Gets the gripper width.

Note

Since the PWM servo has no position feedback, the latest position command is returned.

Parameters

width – width in m.

Returns

err_type_t::OK

virtual void **setReduction**(float reduction)

Manually set reduction.

Parameters

reduction –

virtual void **setOffset**(float offset)

Manually set offset.

Private Members

std::chrono::_V2::system_clock::time_point **_new_cmd_time** = std::chrono::_V2::system_clock::time_point()

class **BaseJoint**

#include <mBaseJoint.h> Generic base class to control a single joint.

This class is a wrapper function to interact with a robot joint either through a *MockJoint* or hardware *Joint* object.

Subclassed by *bioscara_hardware_drivers::Joint*, *bioscara_hardware_drivers::MockJoint*

Public Types

enum **stp_reg_t**

register and command definitions

a register can be read (R) or written (W), each register has a size in bytes. The payload can be split into multiple values or just be a single value. Note that not all functions are implemented.

Values:

enumerator **NONE**

Used for signalling purposes.

enumerator **PING**

R; Size: 1; [(char) ACK].

enumerator **SETUP**

W; Size: 2; [(uint8) holdCurrent, (uint8) driveCurrent].

enumerator **SETRPM**

W; Size: 4; [(float) RPM].

enumerator **GETDRIVERRPM**

enumerator **MOVESTEPS**

W; Size: 4; [(int32) steps].

enumerator **MOVEANGLE**

enumerator **MOVETOANGLE**

W; Size: 4; [(float) degrees].

enumerator **GETMOTORSTATE**

enumerator **RUNCOTINOUS**

enumerator **ANGLEMOVED**

R; Size: 4; [(float) degrees].

enumerator **SETCURRENT**

W; Size: 1; [(uint8) driveCurrent].

enumerator **SETHOLDCURRENT**

W; Size: 1; [(uint8) holdCurrent].

enumerator **SETMAXACCELERATION**

enumerator **SETMAXDECELERATION**

enumerator **SETMAXVELOCITY**

enumerator **ENABLESTALLGUARD**

W; Size: 1; [(uint8) threshold].

enumerator **DISABLESTALLGUARD**

enumerator **CLEARSTALL**

enumerator **SETBRAKEMODE**

W; Size: 1; [(uint8) mode].

enumerator **ENABLEPID**

enumerator **DISABLEPID**

enumerator **ENABLECLOSEDLOOP**

enumerator **DISABLECLOSEDLOOP**

W; Size: 1; [(uint8) 0].

enumerator **SETCONTROLTHRESHOLD**

enumerator **MOVETOEND**

enumerator **STOP**

W; Size: 1; [(uint8) mode].

enumerator **GETPIDERROR**

enumerator **CHECKORIENTATION**

W; Size: 4; [(float) degrees].

enumerator **GETENCODERRPM**

R; Size: 4; [(float) RPM].

enumerator **HOME**

W; Size: 4; [(uint8) current, (int8) sensitivity, (uint8) speed, (uint8) direction].

enumerator **HOMEOFFSET**

R/W; Size: 4; [(float) -].

Public Functions

BaseJoint(const std::string name)

Create a *Joint* object.

The *Joint* object represents a single joint.

Parameters

name – string device name for logging.

~BaseJoint(void)

Destroy the *BaseJoint* object.

Invokes the *disable()* and *deinit()* method to clean up.

virtual *err_type_t* init(void)

Initialize the joint communication.

Returns

err_type_t

virtual *err_type_t* deinit(void)

Denitalize the joint communication.

Returns

err_type_t

virtual *err_type_t* enable(u_int8_t driveCurrent, u_int8_t holdCurrent)

Engages the joint.

This function prepares the motor for movement. After successfull execution the joint is ready to accept *setPosition()* and *setVelocity()* commands.

The function sets the drive and hold current for the specified joint and engages the motor. The currents are in percent of driver max. output (2.5A, check with TMC5130 datasheet or Ustepper documentation)

Parameters

- **driveCurrent** – drive current in 0-100 % of 2.5A output (check uStepper doc.)
- **holdCurrent** – hold current in 0-100 % of 2.5A output (check uStepper doc.)

Returns

err_type_t

virtual *err_type_t* disable(void)

disenganges the joint

invokes *stop()*, sets hold and drive current to 0 and sets the the joint brake mode to freewheeling

Returns

err_type_t

virtual *err_type_t* home(float velocity, u_int8_t sensitivity, u_int8_t current)

Blocking implementation to home the joint.

Homing the joint is neccessary after its controller has been powered off. The joint can be moved while the encoders are inactive and hence the position at startup is unknow. See *_home()* for information on implementation and description of the paramters.

This is a blocking implementation which only returns after the the joint is no longer busy. First *startHoming()* is called, and subsequently waits with *wait_while_busy()* to finish homing. Lastly *postHoming()* is called.

Returns

err_type_t

virtual *err_type_t* startHoming(float velocity, u_int8_t sensitivity, u_int8_t current)

non-blocking implementation to home the joint

Homing the joint is neccessary after its controller has been powered off. The joint can be moved while the encoders are inactive and hence the position at startup is unknow. See *_home()* for information on implementation and description of the paramters.

This method returns immediatly after starting the homing sequence and should be used when the blocking implementation is not acceptable, for example in a realtime loop.

This method sets the current_b_cmd flag to *stp_reg_t::HOME*

Returns

err_type_t

virtual *err_type_t* **postHoming**(void)

perform tasks after a non-blocking homing

This method resets the current_b_cmd to *stp_reg_t::NONE*, checks if the joint is homed, and saves the homing offset to the joint.

Returns

err_type_t

virtual *err_type_t* **getPosition**(float &pos) = 0

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Warning

If the joint is not homed this method does not return an error. Instead pos will be 0.0.

Parameters

pos – the current joint position in rad or m.

Returns

err_type_t

virtual *err_type_t* **setPosition**(float pos)

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Parameters

pos – the commanded joint position in rad or m.

Returns

err_type_t

virtual *err_type_t* **moveSteps**(int32_t steps)

Move full steps.

This function can be called even when not homed.

Parameters

steps – number of full steps

Returns

err_type_t

virtual *err_type_t* **getVelocity**(float &vel) = 0

get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

vel – the current joint velocity in rad/s or m/s.

Returns

err_type_t

virtual *err_type_t* **setVelocity**(float vel)

Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

vel – the commanded joint velocity in rad/s or m/s.

Returns

err_type_t

virtual *err_type_t* **checkOrientation**(float angle = 2.0)

Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.

Note

This is a blocking function.

Parameters

angle – degrees how much the motor should turn. A few degrees is sufficient.

Returns

err_type_t

virtual *err_type_t* **stop**(void)

Stops the motor.

Stops the motor by setting the maximum velocity to zero and the position setpoint to the current position

Returns

err_type_t

virtual *err_type_t* **disableCL**(void)

virtual *err_type_t* **setDriveCurrent**(u_int8_t current)

Set the Drive Current.

Parameters

current – 0% - 100% of driver current

Returns

err_type_t

virtual *err_type_t* **setHoldCurrent**(u_int8_t current)

Set the Hold Current.

Parameters

current – 0% - 100% of driver current

Returns

err_type_t -1 on communication error,

virtual *err_type_t* **setBrakeMode**(u_int8_t mode)

Set Brake Mode.

Parameters

mode – Freewheel: 0, Coolbrake: 1, Hardbrake: 2

Returns

err_type_t

virtual *err_type_t* **setMaxAcceleration**(float maxAccel)

Set the maximum permitted joint acceleration (and deceleration) in rad/s² or m/s² for cylindrical and prismatic joints respectively.

Parameters

maxAccel – maximum joint acceleration.

Returns

err_type_t

virtual *err_type_t* **setMaxVelocity**(float maxVel)

Set the maximum permitted joint velocity in rad/s or m/s for cylindrical and prismatic joints respectively.

Parameters

maxVel – maximum joint velocity.

Returns

err_type_t

virtual *err_type_t* **enableStallguard**(u_int8_t sensitivity)

Enable encoder stall detection of the joint.

If the PID error exceeds the set threshold a stall is triggered and the motor disabled. A detected stall can be reset by homing or reenabling the joint using *enable()*.

Note

If stall detection shall be enabled, invoke this method AFTER enabling the joint with *enable()*.

Parameters

sensitivity – value of threshold. 0 - 255 where lower is more sensitive.

Returns

err_type_t

virtual bool **isHomed**(void)

Checks the state if the motor is homed.

Reads the internal state flags from the last transmission. If an update is necessary call *getFlags()* before invoking this function.

Returns

true if the motor is homed, false if not.

virtual bool **isEnabled**(void)

Checks the state if the motor is enabled.

Reads the internal state flags from the last transmission. If an update is necessary call *getFlags()* before invoking this function. If the motor actually can move depends on the state of the STALLED flag which can be checked using *Joint::isStalled()*.

Returns

true if the motor is enabled, false if not.

virtual bool **isStalled**(void)

Checks if the motor is stalled.

Reads the internal state flags from the last transmission. If an update is necessary call *getFlags()* before invoking this function.

Returns

true if the motor is stalled, false if not.

virtual bool **isBusy**(void)

Checks if the joint controller is busy processing a blocking command.

Reads the internal state flags from the last transmission. If an update is necessary call *getFlags()* before invoking this function.

Returns

true if a blocking command is currently executing, false if not.

virtual *err_type_t* **getFlags**(u_int8_t &flags)

get the latest driver state flags from the joint

Parameters

flags – if succesfull, populated with the latest flags

Returns

err_type_t

virtual *err_type_t* **getFlags**(void)

Overload of *getFlags(u_int8_t &flags)*

Returns

err_type_t

virtual *stp_reg_t* **getCurrentBCmd**(void)

get the currently active blocking command

Returns

The the command of type *stp_reg_t*

Public Members

std::string **name**

Joint name for logging.

class **Gripper** : public bioscara_hardware_drivers::*BaseGripper*

#include <mGripper.h> Child class implementing control of the hardware gripper.

Use this class if you wish to control the actual hardware by generating a PWM voltage on GPIO18 of the Raspberry Pi.

Public Functions

Gripper(float reduction, float offset, float min, float max, float backup_init_pos)

Construct a new *Gripper* object.

see the *BaseGripper* constructor for a description of the parameters.

virtual *err_type_t* **enable**(void) override

Activates the the hardware.

Invokes *BaseGripper::enable()* and starts the PWM generation but does not set a position. Must be called before a position is set. The PWM pin is GPIO18 on the Raspberry Pi 4. PWM chip is 0, channel 0.

Returns

err_type_t::OK on success, *err_type_t::COMM_ERROR* if PWM can not be enabled

virtual *err_type_t* **disable**(void) override

Deactivates the hardware.

Invokes *BaseGripper::disable()* and stops the servo by disabling the PWM generation.

Returns

err_type_t::OK

virtual *err_type_t* **setPosition**(float width) override

Sets the gripper width in m.

Invokes *BaseGripper::setPosition()*, transforms the desired gripper width to a servo angle using the JOINT2ACTUATOR() macro and finally sets the servo position using setServoPosition()

Parameters

width – width in m.

Returns

see setServoPosition()

class **Joint** : public bioscara_hardware_drivers::*BaseJoint*

#include <mJoint.h> Representing a single hardware joint connected via I2C.

Public Functions

Joint(const std::string name, const int address, const float reduction, const float min, const float max)

Create a *Joint* object.

The *Joint* object represents a single joint and its actuator. Each *Joint* has a transmission with the following relationship:

$$\text{actuator position} = (\text{joint position} - \text{offset}) * \text{reduction}$$

$$\text{joint position} = \text{actuator position} / \text{reduction} + \text{offset}$$

Parameters

- **name** – string device name for identification
- **address** – 1-byte I2C device address (0x11 ... 0x14) for J1 ... J4
- **reduction** – gear reduction of the joint. This is used to transform position and velocity values between in joint units and actuator (stepper) units. The sign depends on the direction the motor is mounted and is turning. Adjust such that the joint moves in the positive direction on on positive joint commands. Cable polarity has no effect since the motors automatically adjust to always run in the ‘right’ direction from their point of view.

J1: 35

J2: $-2 * \pi / 0.004$ (4 mm linear movement per stepper revolution)

J3: 24

J4: 12

- **min** – lower joint limit in joint units.

J1: -3.04647

J2: -0.0016

J3: -2.62672

J4: -3.01069

- **max** – upper joint limit in joint units.

J1: 3.04647

J2: 0.3380

J3: 2.62672

J4: 3.01069

~Joint(void)

Destroy the *Joint* object.

Create a *Joint* object.

virtual *err_type_t* **init**(void) override

Initialize connection to a joint via I2C.

Adds the joint to the I2C bus and tests if is responsive by invoking `checkCom()`. If the joint is homed, retrieve the homing position stored on the joint by invoking `getHomingOffset()`.

Returns

err_type_t

virtual *err_type_t* **deinit**(void) override

Disconnects from a joint.

Removes the joint from the I2C bus by invoking `closeI2CDevHandle()`.

Returns

err_type_t

virtual *err_type_t* **enable**(u_int8_t driveCurrent, u_int8_t holdCurrent) override

Engages the joint.

This function prepares the motor for movement. After successful execution the joint is ready to accept *setPosition()* and *setVelocity()* commands.

The function sets the drive and hold current for the specified joint and engages the motor. The currents are in percent of driver max. output (2.5A, check with TMC5130 datasheet or Ustepper documentation)

Parameters

- **driveCurrent** – drive current in 0-100 % of 2.5A output (check uStepper doc.)
- **holdCurrent** – hold current in 0-100 % of 2.5A output (check uStepper doc.)

Returns

err_type_t

virtual *err_type_t* **postHoming**(void) override

perform tasks after a non-blocking homing

Invokes *BaseJoint::postHoming()* and additionally saves the homing offset to the joint controller `setHomingOffset()`

Returns

err_type_t

virtual *err_type_t* **getPosition**(float &pos) override

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Warning

If the joint is not homed this method does not return an error. Instead `pos` will be 0.0.

Parameters

pos – the current joint position in rad or m.

Returns

`err_type_t`

virtual `err_type_t` **setPosition**(float pos) override

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Parameters

pos – the commanded joint position in rad or m.

Returns

`err_type_t`

virtual `err_type_t` **moveSteps**(int32_t steps) override

Move full steps.

This function can be called even when not homed.

Parameters

steps – number of full steps

Returns

`err_type_t`

virtual `err_type_t` **getVelocity**(float &vel) override

get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

vel – the current joint velocity in rad/s or m/s.

Returns

`err_type_t`

virtual `err_type_t` **setVelocity**(float vel) override

Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

vel – the commanded joint velocity in rad/s or m/s.

Returns

`err_type_t`

virtual `err_type_t` **checkOrientation**(float angle = 2.0) override

Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.

Note

This is a blocking function.

Parameters

angle – degrees how much the motor should turn. A few degrees is sufficient.

Returns

err_type_t

virtual *err_type_t* **stop**(void) override

Stops the motor.

Stops the motor by setting the maximum velocity to zero and the position setpoint to the current position

Returns

err_type_t

virtual *err_type_t* **disableCL**(void) override

virtual *err_type_t* **setDriveCurrent**(u_int8_t current) override

Set the Drive Current.

Parameters

current – 0% - 100% of driver current

Returns

err_type_t

virtual *err_type_t* **setHoldCurrent**(u_int8_t current) override

Set the Hold Current.

Parameters

current – 0% - 100% of driver current

Returns

err_type_t -1 on communication error,

virtual *err_type_t* **setBrakeMode**(u_int8_t mode) override

Set Brake Mode.

Parameters

mode – Freewheel: 0, Coolbrake: 1, Hardbrake: 2

Returns

err_type_t

virtual *err_type_t* **setMaxAcceleration**(float maxAccel) override

Set the maximum permitted joint acceleration (and deceleration) in rad/s² or m/s² for cylindrical and prismatic joints respectively.

Parameters

maxAccel – maximum joint acceleration.

Returns

err_type_t

virtual *err_type_t* **setMaxVelocity**(float maxVel) override

Set the maximum permitted joint velocity in rad/s or m/s for cylindrical and prismatic joints respectively.

Parameters

maxVel – maximum joint velocity.

Returns

err_type_t

virtual *err_type_t* **enableStallguard**(u_int8_t sensitivity) override

Enable encoder stall detection of the joint.

If the PID error exceeds the set threshold a stall is triggered and the motor disabled. A detected stall can be reset by homing or reenabling the joint using *enable()*.

Note

If stall detection shall be enabled, invoke this method AFTER enabling the joint with *enable()*.

Parameters

sensitivity – value of threshold. 0 - 255 where lower is more sensitive.

Returns

err_type_t

virtual *err_type_t* **getFlags**(void) override

Overload of *getFlags(u_int8_t &flags)*

Returns

err_type_t

Private Functions

template<typename T>

int **read**(const stp_reg_t reg, T &data, u_int8_t &flags)

Wrapper function to request data from the I2C slave.

Allocates a buffer of size sizeof(T) + RFLAGS_SIZE. Invokes readFromI2CDev(), and copies the received payload to *data* and the transmission flags to *flags*. See BaseJoint::flags for details.

Template Parameters

T – Datatype of value to be transmitted

Parameters

- **reg** – stp_reg_t register to read
- **data** – reference to store payload.
- **flags** – reference to a byte which stores the return flags

Returns

0 on OK, negative on error

template<typename T>

int **write**(const stp_reg_t reg, T data, u_int8_t &flags)

Wrapper function to send command to the I2C slave.

Allocates a buffer of size sizeof(T) + RFLAGS_SIZE. Copies *data* to the buffer and invokes writeToI2CDev(). The flags received from the transaction are copied to *flags*. The flags are described in *Joint::read()*.

Template Parameters

T – Datatype of value to be transmitted

Parameters

- **reg** – stp_reg_t command to execute

- **data** – payload to transmit. It is the users responsibility to populate the right amount of data for the relevant register
- **flags** – reference to a byte which stores the return flags

Returns

0 on OK, negative on error

Private Members

int **address**

I2C adress.

int **handle** = -1

I2C bus handle.

class **MockGripper** : public bioscara_hardware_drivers::*BaseGripper*

#include <mMockGripper.h> Child class for to mock the gripper hardware.

Use this class if you dont wish to use the gripper hardware. Commands are simply mirrored.

Public Functions

MockGripper(float reduction, float offset, float min, float max, float backup_init_pos)

Construct a new *MockGripper* object.

see the *BaseGripper* constructor for a description of the parameters.

class **MockJoint** : public bioscara_hardware_drivers::*BaseJoint*

#include <mMockJoint.h> Representing a single joint mocking the hardware joint.

This mock hardware implementation simply mirrors (potentially delayed) commands.

Note

Much of this implementation is very basic not well written.

Public Functions

MockJoint(const std::string name)

virtual *err_type_t* **enable**(u_int8_t driveCurrent, u_int8_t holdCurrent) override

Engages the joint.

This function prepares the motor for movement. After successfull execution the joint is ready to accept *setPosition()* and *setVelocity()* commands.

The function sets the drive and hold current for the specified joint and engages the motor. The currents are in percent of driver max. output (2.5A, check with TMC5130 datasheet or Ustepper documentation)

Parameters

- **driveCurrent** – drive current in 0-100 % of 2.5A output (check uStepper doc.)
- **holdCurrent** – hold current in 0-100 % of 2.5A output (check uStepper doc.)

Returns

err_type_t

virtual *err_type_t* **disable**(void) override

disengages the joint

invokes *stop()*, sets hold and drive current to 0 and sets the the joint brake mode to freewheeling**Returns**

err_type_t

virtual *err_type_t* **getPosition**(float &pos) override

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

WarningIf the joint is not homed this method does not return an error. Instead *pos* will be 0.0.**Parameters****pos** – the current joint position in rad or m.**Returns**

err_type_t

virtual *err_type_t* **setPosition**(float pos) override

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Parameters**pos** – the commanded joint position in rad or m.**Returns**

err_type_t

virtual *err_type_t* **getVelocity**(float &vel) override

get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters**vel** – the current joint velocity in rad/s or m/s.**Returns**

err_type_t

virtual *err_type_t* **setVelocity**(float vel) override

Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters**vel** – the commanded joint velocity in rad/s or m/s.**Returns**

err_type_t

virtual *err_type_t* **checkOrientation**(float angle = 2.0) override

Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.

Note

This is a blocking function.

Parameters

angle – degrees how much the motor should turn. A few degrees is sufficient.

Returns

`err_type_t`

virtual `err_type_t` **stop**(void) override

Stops the motor.

Stops the motor by setting the maximum velocity to zero and the position setpoint to the current position

Returns

`err_type_t`

virtual `err_type_t` **getFlags**(void) override

Overload of `getFlags(u_int8_t &flags)`

Returns

`err_type_t`

virtual bool **isHomed**(void) override

Checks the state if the motor is homed.

Reads the internal state flags from the last transmission. If an update is necessary call `getFlags()` before invoking this function.

Returns

true if the motor is homed, false if not.

Private Functions

float **getDeltaT**(std::chrono::_V2::system_clock::time_point &last_call, bool update = true)

Compute the time since the given time point.

Parameters

- **last_call** – the time point to calculate the duration
- **update** – If true compute the duration and also reset *last_call* to current time

Returns

elapsed time in seconds since *last_call*

Private Members

float **q** = 0.0

position

float **qd** = 0.0

velocity

```
std::chrono::_V2::system_clock::time_point last_set_position =
std::chrono::high_resolution_clock::now()

std::chrono::_V2::system_clock::time_point last_set_velocity = last_set_position

std::chrono::_V2::system_clock::time_point async_start_time = last_set_position

stp_reg_t op_mode = NONE
```

1.2 User Guides

1.2.1 Installing

1.2.2 Building the ROS2 Workspace

After importing all dependencies the ROS2 workspace *lib/ros2_ws/* will contain a lot of packages, primarily MoveIt and MTC packages along the usual Bioscara packages. In order to be able to run them, they all have to be built. *colcon* is the ROS2 building tool that is used for this purpose.

Note

This guide assumes that you are in the *lib/ros2_ws/* workspace. If you are not, navigate to it:

```
cd lib/ros2_ws/
```

Building the Entire Workspace including MoveIt2 and MTC

Both MoveIt2 and the MTC are huge packages that will make the Raspberry Pi crash under the load. Even for normal computers there is a high chance of freezing. For this reason the *colcon build* command is invoked with very restrictive parameters to allow the compilation to finish at all. The build command that should be used to compile the entire workspace at once is:

```
MAKEFLAGS="-j1 -l1" colcon build --mixin release --executor sequential --symlink-install_
↪--continue-on-error > log.out &
```

```
disown
```

The *MAKEFLAGS="-j1 -l1"* restrict the process to a single core, the *--executor sequential* flag ensures that only one package at the time is built (otherwise up to 8 would be built), the *--continue-on-error* flag will try to keep the compilation going if one package fails. This is very important since the compilation can take up to 10 (!) hours on the Raspberry Pi 4B with 4GB memory.

The high memory usage will cause the Raspberry Pi to become unresponsive at times, potentially dropping the SSH connection and thus the user session that started the build proces. To avoid this the *colcon* output is being piped *> log.out* to a file and the process is being sent to the background *&*. Before disconnecting from the session the process must be disowned by calling *disown*. The build job will now continue even if the user session that started it terminates.

Building selected Packages

Luckil as long as you dont modify any of the MoveIt or MTC packages or accidentally delete the *build/* or *install/* directories, you will not have to build the entire workspace again. To rebuild a selected set of packages simply invoke *colcon* like this:

```
colcon build --mixin release --symlink-install --packages-select <package_name> <package_
↪name> <package_name>
```

or select the by a regex expression `--packages-select-regex <package_name_regex>` or any other (selection flag)[<https://colcon.readthedocs.io/en/released/reference/package-selection-arguments.html>]

Other Notes

Using the `--symlink-install` flag create symlinks instead of copying files from the source and build directories where possible. This way it is not necessary to recompile a package if a non-C++ file has been modified. Changing parameter yaml-files, Pthyon scripts or URDF files is thus simpler and faster.

The `--mixin release` makes the packages compile in release mode, which is very important for performance reasons. Not using this flag will make path planning very slow.

If compilation fails, in particular if a package's *CMakeLists.txt* has been modified, it can help to selectively clean the *build/* and *install/* directory:

```
rm -rf build/<package_name> install/<package_name>
```

1.2.3 Getting Started

1.2.4 Building the Documentation locally

It is recommended to use a virtual environment to manage all dependencies. All following commands are executed from the `~/bioscara/docs/` directory:

```
cd docs
```

Create the Virtual Environment and Install Dependencies

These steps need to be executed only once at the first time. Create the virtual python environment:

```
python -m venv .venv
```

Before installing the dependencies the environment needs to be activated:

```
source .venv/bin/activate
```

The we can install the required packages to build the documentation:

```
pip install -r requirements-docs.txt
```

This installs Sphinx and its packages.

Additionally install doxygen

```
sudo apt-get update
sudo apt-get install doxygen
```

And install all LaTeX tools:

```
sudo apt-get install texlive texlive-font-utils texlive-fonts-recommended texlive-latex-
↳extra latexmk
```

Building the Documentation

First make sure the virtual environment is activated:

```
source .venv/bin/activate
```

Then run doxygen from the *docs/doxygen/* directory to produce HTML, XML and LaTeX output:

```
cd doxygen
doxygen
```

Compile the LaTeX to a PDF:

```
cd doxygen/latex
make
```

Finally build the Sphinx documentation:

```
sphinx-build -b html . build/html
```

It can also be compiled to a PDF:

```
sphinx-build -M latexpdf . build/latex
```

1.2.5 Modifying and Flashing the Joint Firmware

The joint firmware is programmed in C++ using the Arduino framework since the uStepperS32 stepper library is released for that ecosystem.

Prerequisites

In order to flash firmware to the uStepperS32 stepper controllers some prerequisites must be met. The following guide is copied for reference from the [uStepper website](#).

Install the Arduino IDE

Installing the uStepperS32 library and flashing the firmware is done through the Arduino IDE which needs to be installed on the developer's computer from [here](#).

Install STM32 CUBE Programmer

The uStepper is based on an STM32F401 microcontroller. The STM32 CUBE Programmer is used behind the scenes to flash the firmware from the Arduino IDE. Install the programmer from [here](#).

Add the uStepperS32 library to the Arduino IDE

In the Arduino IDE, go to *File->preferences* and append the URL <https://raw.githubusercontent.com/uStepper/uStepperSTM32Hardware/master/package.json> to the Additional Boards Manager URLs field at the bottom and accept with OK.

Then install the uStepper board through the board manager under *Tools->Board->Boards Manager*. Search for “ustep-” and the board should show up after the board manager has refreshed. Then press INSTALL.

Finally it should now be possible to select the uStepper STM32 boards under *tools->boards*.

Install the uStepperS32 Library

Go to *Sketch->Include Library->Manage Libraries*, search for “uStepperS32”, and press INSTALL. Now both the hardware and the library are setup for compilation and flashing.

Flashing the Firmware

To flash the firmware on the uStepper board, open the *lib/joint_firmware/joint/joint.ino* firmware file with the Arduino IDE, connect the uStepper board with a USB-C cable to the computer, select the correct serial port and the board as “uStepper STM32 boards” under the “board and port” tab.

The follow the steps to flash the firmware:

1. Press and hold the BOOT button on the uStepper board
2. Keep the BOOT button is pressed and press the RESET button for at least 0.5 s
3. Release the BOOT button
4. Select UPLOAD in the Arduino IDE. The board is now in download mode and will load the new firmware.

Monitor the Arduino IDE output console for potential error messages.

Other Notes

The buttons on the uStepper board are very difficult to reach on J2 if it is mounted on the robot. A thin non-conductive stick can be used to press the BOOT button, the RESET button should be reachable by hand.

1.2.6 Creating a IKFast Inverse Kinematics Plugin

Important

This guide can only serve as a starting point since the desired result could not be achieved!

The IKFast is an analytical inverse kinematics solver that automatically generates a robots IK solver code based on its URDF file. This guide documents which steps have been undertaken to generate the IKFast solver plugin for MoveIt.

The result however has always been that the robot did not move when dragging on its handles in MoveIt.

Background

Please refer to the original guide on the MoveIt documentation page [here](#) for the information about the individual commands.

Plugin Creation

make sure Docker is installed:

```
docker -v
```

If that is not the case install Docker by following the guide [here](#).

if that is done, make sure your user is added to the “docker” group:

```
sudo usermod -aG docker ${USER}
```


log out and back in to apply the new group:

```
su -l ${USER}
```

navigate to the `lib/ros2_ws/src/dalsa_bioscara_arm` directory:

```
cd lib/ros2_ws/src/dalsa_bioscara_arm
```

export the arm as an ENV variable:

```
export MYROBOT_NAME="bioscara_arm"
```

Create a urdf file from the `scene.xacro` which will be parsed by the plugin generation script.

```
ros2 run xacro xacro -o $MYROBOT_NAME.urdf bioscara_arm_description/urdf/scene.xacro
```

Then create then call the plugin creation script:

```
ros2 run moveit_kinematics auto_create_ikfast_moveit_plugin.sh --iktype <type> $MYROBOT_
NAME.urdf arm base_link tool_flange
```

On first run this will take a while to pull and build the docker images.

Select one 4-DOF <type> from the following ([source](#)):

- **Transform6D** - end effector reaches desired 6D transformation
- **Rotation3D** - end effector reaches desired 3D rotation
- **Translation3D** - end effector origin reaches desired 3D translation
- **Direction3D** - direction on end effector coordinate system reaches desired direction
- **Ray4D** - ray on end effector coordinate system reaches desired global ray
- **Lookat3D** - direction on end effector coordinate system points to desired 3D position
- **TranslationDirection5D** - end effector origin and direction reaches desired 3D translation and direction. Can be thought of as Ray IK where the origin of the ray must coincide.
- **TranslationXY2D** - end effector origin reaches desired XY translation position, Z is ignored. The coordinate system with relative to the base link.
- **TranslationLocalGlobal6D** - local point on end effector origin reaches desired 3D global point. Because both local point and global point can be specified, there are 6 values.
- **TranslationXAxisAngle4D**, **TranslationYAxisAngle4D**, **TranslationZAxisAngle4D** - end effector origin reaches desired 3D translation, manipulator direction makes a specific angle with x/y/z-axis (defined in the manipulator base link's coordinate system)
- **TranslationXAxisAngleZNorm4D**, **TranslationYAxisAngleXNorm4D**, **TranslationZAxisAngleYNorm4D** - end effector origin reaches desired 3D translation, manipulator direction needs to be orthogonal to z, x, or y axis and be rotated at a certain angle starting from the x, y, or z axis (defined in the manipulator base link's coordinate system)

None of the last 6 types have yielded a successful result yet.

Build the new Plugin Package

Navigate to the root of the workspace:

```
cd lib/ros2_ws
```

and build the new package:

```
colcon build --symlink-install --mixin release --packages-select bioscara_arm_ikfast_  
↪ plugin
```

Change the kinematics.yaml

Almost done, change the `lib/ros2_ws/src/moveit_configurations/bioscara_arm_gripper_128_moveit_config/config/kinematics.yaml` file to the new IK solver:

```
arm:  
  kinematics_solver: bioscara_arm/IKFastKinematicsPlugin  
  ...
```

Try it out

Start the robot (mock or hardware), `move_group` and `rviz` as described in the [guide](#).

INDEX

B

bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper (C++ function), 23
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper (C++ class), 15
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::new_cmd_time (C++ member), 17
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::BaseGripper (C++ function), 16
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::BaseGripper (C++ function), 15
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::deinit (C++ function), 16
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::disable (C++ function), 16
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::enable (C++ function), 16
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::getPosition (C++ function), 17
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::init (C++ function), 16
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::setOffset (C++ function), 17
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::setPosition (C++ function), 16
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseGripper::setReduction (C++ function), 17
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint (C++ class), 17
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::BaseJoint (C++ function), 19
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::BaseJoint (C++ function), 19
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::checkOrientation (C++ function), 22
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::deinit (C++ function), 20
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::disable (C++ function), 20
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::disableCL (C++ function), 22
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::enable (C++ function), 20
 bioscara_hardware_drivers::bioscara_hardware_drivers::BaseJoint::enableStallguard (C++ function), 23

Index	41
--------------	-----------

(C++ member), 10
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 7
 (C++ member), 11
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 4
 (C++ member), 9
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ class), 11
 (C++ member), 10
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 11
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 10
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 10
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 9
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ struct), 14
 (C++ member), 10
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ struct), 11
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 11
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 11
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 11
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ member), 14
 (C++ member), 11
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 12
 (C++ member), 9
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 12
 (C++ function), 3
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 12
 (C++ function), 3
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 12
 (C++ function), 3
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 13
 (C++ function), 4
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 12
 (C++ function), 6
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 12
 (C++ function), 2
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 13
 (C++ function), 3
 bioscara_hardware_interfaces::bioscara_hardware_interfaces (C++ function), 13
 (C++ function), 6
 bioscara_hardware_interfaces::bioscara_hardware_interfaces::BioscaraArmHardwareInterface::perform_command (C++ member), 1
 bioscara_hardware_interfaces::bioscara_hardware_interfaces::BioscaraArmHardwareInterface::prepare_command (C++ function), 5
 bioscara_hardware_interfaces::bioscara_hardware_interfaces::BioscaraArmHardwareInterface::read (C++ function), 4
 bioscara_hardware_interfaces::bioscara_hardware_interfaces::BioscaraArmHardwareInterface::split_interfaces (C++ function), 7
 bioscara_hardware_interfaces::bioscara_hardware_interfaces::BioscaraArmHardwareInterface::start_homing (C++ function), 7
 bioscara_hardware_interfaces::bioscara_hardware_interfaces::BioscaraArmHardwareInterface::stop_homing